

Early Warning Systems for Banking Crises

Evaluating the Interpretability–Performance Trade-off
Under Structural Non-Stationarity

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10-min Presentation

The Problem: A Known Failure, Unknown Mechanism

What the literature established — and what was missing

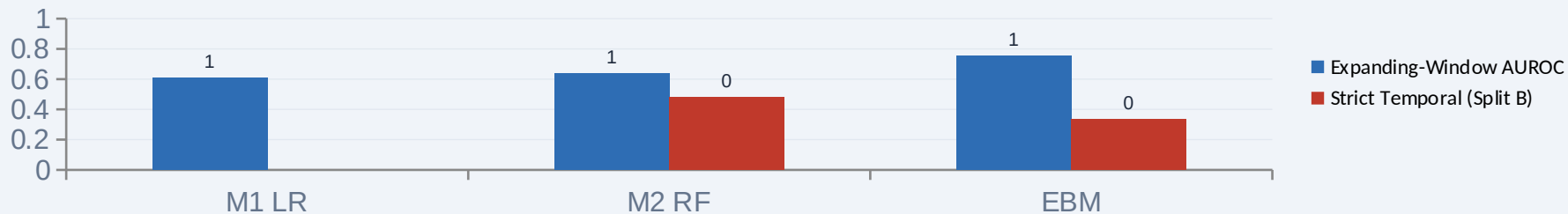
What the literature knew

- ML (RF, XGBoost) outperforms logistic regression under expanding-window CV
- Strict temporal validation: all models fall below AUROC 0.75 benchmark (Beutel et al., 2019)
- Extending training data does not resolve failure
- FinBERT sentiment provides directional signal

What was missing

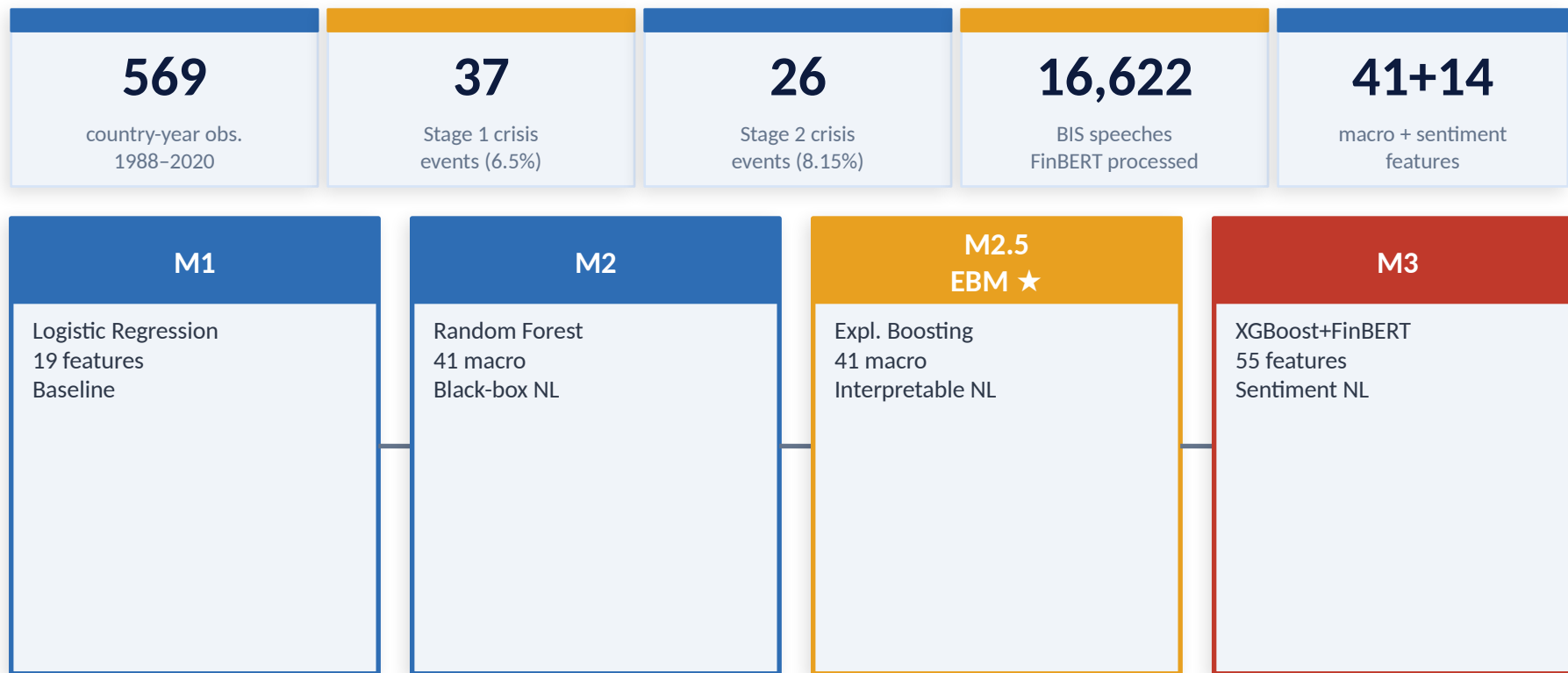
- No mechanism-level explanation of why failure occurs
- No test: are interpretable models more or less vulnerable?
- No clean separation of algorithm vs. sentiment effect
- SHAP is approximate — cannot distinguish model error from attribution error

Expanding-Window vs Strict Temporal AUROC



Data, Design and Model Sequence

JST Macrohistory R6 · 18 economies · 1985–2020

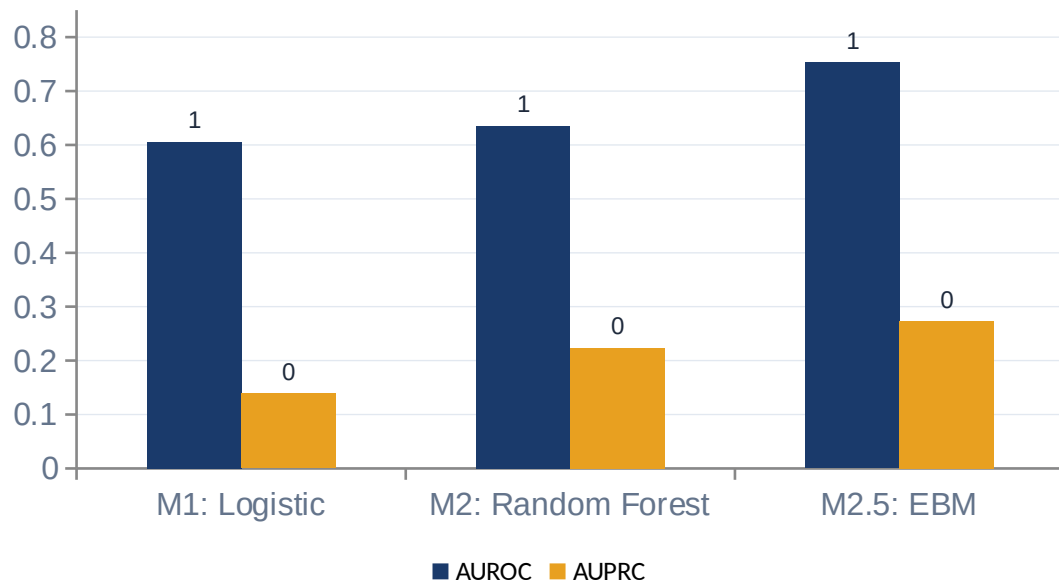


★ Novel contribution: only model that makes failure mechanism directly visible

Stage 1 Results: EBM Leads Across All Metrics

Expanding-window evaluation · 1988–2020 · 37 crisis events · 27 folds

Stage 1 AUROC & AUPRC by Model



Model	AUROC	AUPRC	Brier
M1: Logistic	0.606	0.139	0.157
M2: Random Forest	0.636	0.224	0.060
M2.5: EBM ★	0.753	0.273	0.064

Key observation

EBM achieves AUROC 0.753 — meeting the 0.75 benchmark — with identical features and CV design as M2 RF (AUROC 0.636). The AUROC gap of +0.117 has partially overlapping CIs. This is promising but subject to the same evaluation bias as all expanding-window figures.

All metrics are upper bounds on prospective performance — strict temporal validation in Slide 5 reveals the true picture.

The Prospective Collapse: Why EBM Fails Hardest

Strict temporal validation — the primary empirical finding

M2: Random Forest — Split B

AUROC

0.481

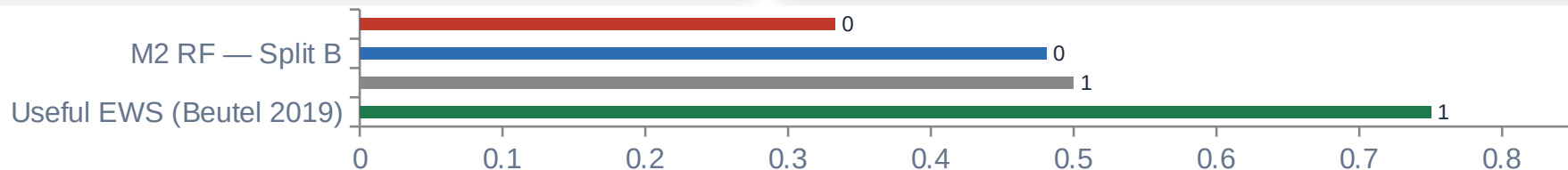
Below random (0.5) · Train 1988–2002 · Test 2003–2011

M2.5: EBM ★ — Split B

AUROC

0.333

Severely below random · Worse than RF · Most flexible = most brittle



Why EBM Fails: Shape Function Regime Instability

Figure 4.7 — direct evidence of non-stationarity in the DGP

INSERT FIGURE 4.7

EBM shape functions:
term spread lags 1-3 & total credit lags 1-3

Blue = Pre-GFC (2003-2006)
Red dashed = GFC-inclusive (2003-2010)

Independent parametric confirmation: Panel logistic — $\text{term_spread_lag1 } \beta = -0.646$ ($p=0.004$) vs $\text{lag3 } \beta = +0.630$ ($p=0.003$). Sign reversal confirmed from a completely separate analytical framework.

Finding 1 — Same feature, opposite risk

At high `term_spread_lag3` values: pre-GFC model assigns REDUCED risk; GFC-inclusive assigns INCREASED risk. The curves cross. Sign reversal at the top-ranked feature.

Finding 2 — Not an approximation error

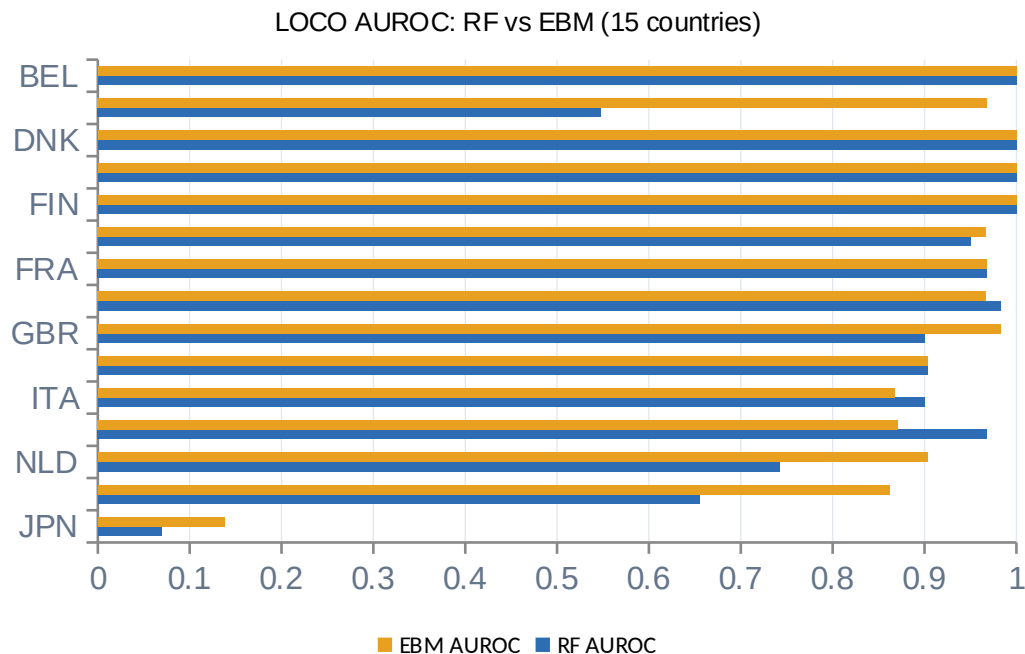
EBM shape functions ARE the model parameters. No SHAP step. Any instability = property of the data-generating process, not the attribution tool. This is the unique contribution.

Finding 3 — Mechanism explains 0.333

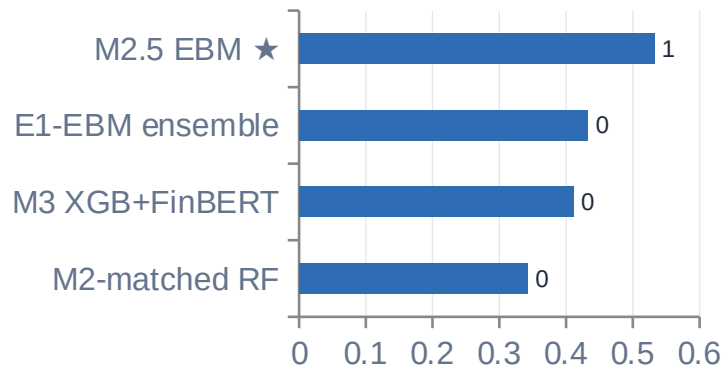
Model trained on pre-GFC applies blue-line mappings to GFC data. Gets the sign wrong on the dominant feature. Systematic misclassification. AUROC below random is the arithmetically necessary result.

Cross-Country and Stage 2 Performance

LOCO validation · Stage 2 model comparison · EBM vs sentiment



Stage 2 AUPRC Comparison



EBM macro-only beats M3+sentiment

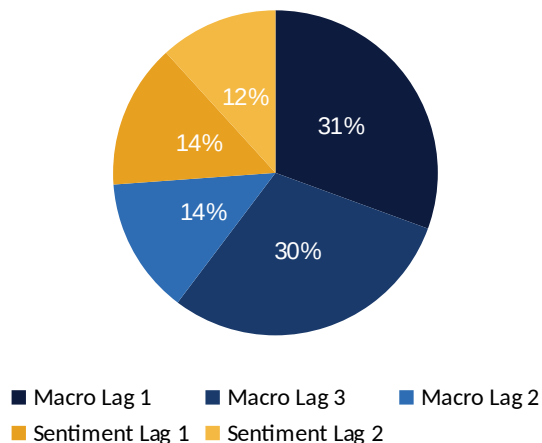
EBM AUPRC 0.532 > M3 (macro+FinBERT) 0.411. Algorithm choice explains at least as much of the Stage 2 difference as the sentiment extension. CI [-0.130, +0.312] spans zero.

LOCO: EBM mean 0.893 (std 0.215) vs RF 0.839 (0.254). Japan near-random for both models — asset-deflation dynamics poorly represented in cross-country training distribution.

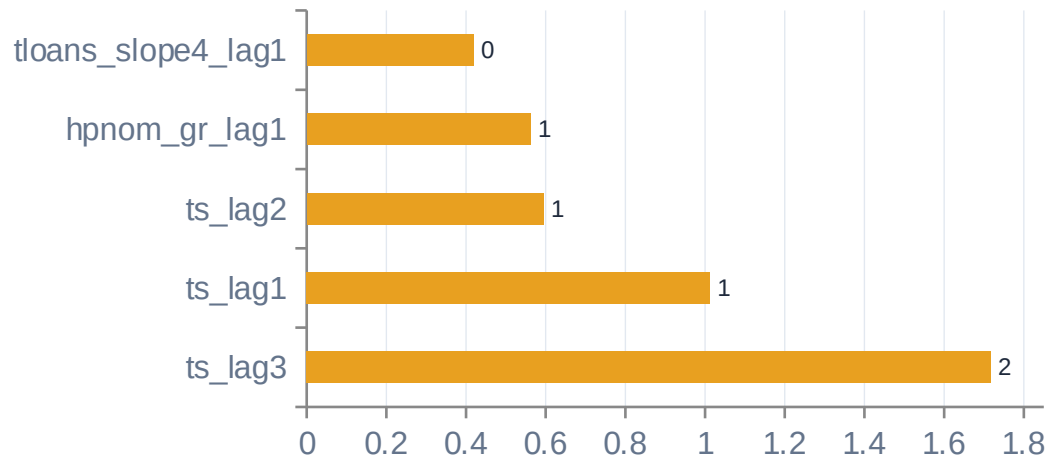
Feature Attribution and Calibration

SHAP vs EBM native importance · ECE and Brier scores

M3 SHAP Attribution by Group



EBM Top-5 Native Feature Importance



Calibration (Stage 2):

Model	ECE	Brier
M2-matched RF	0.127	0.060
M2.5 EBM ★	0.084	0.067
M3 XGB+FinBERT	0.109	0.061
E1-EBM ensemble	0.056*	—

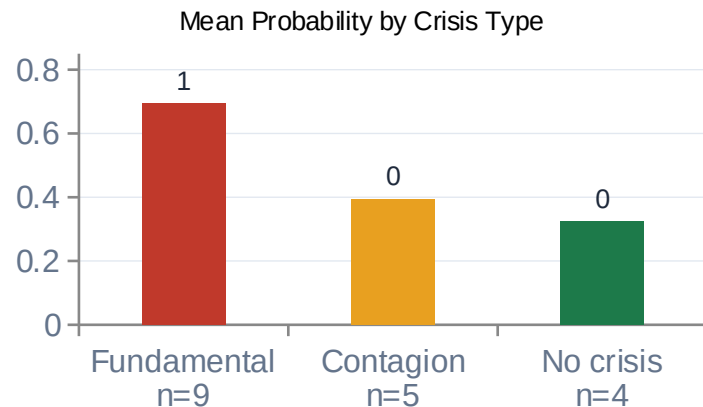
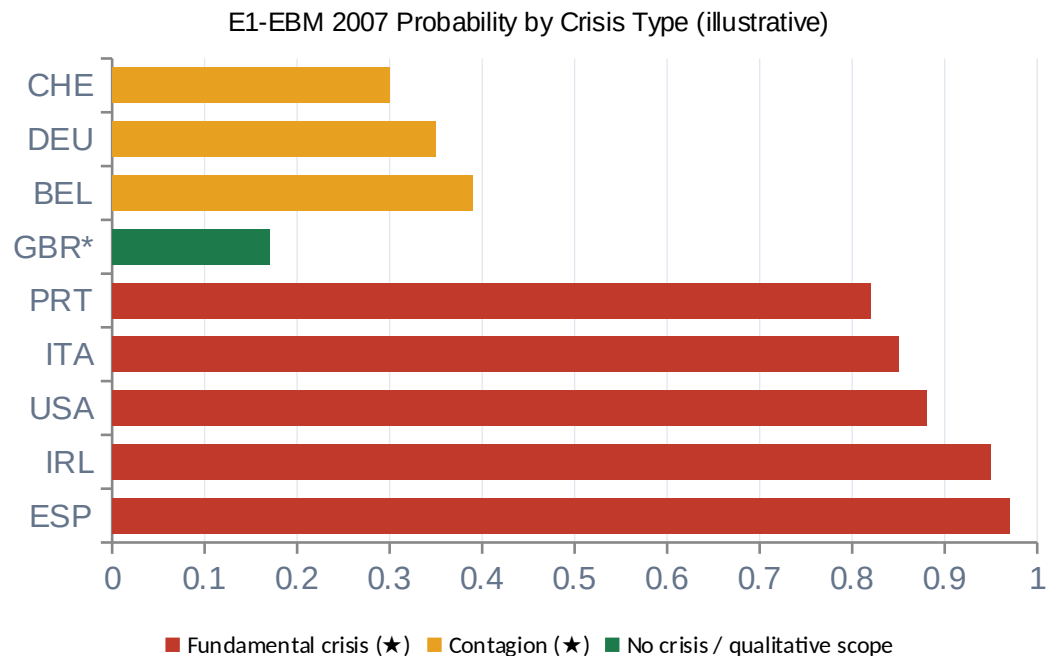
*E1-EBM Brier 0.056 = best calibration of any configuration

Key: macro dominates, EBM best ECE

Both SHAP (73.8% macro) and EBM native importance agree: term_spread_lag3 is the dominant feature. No sentiment in top-5. EBM achieves best ECE (0.084) without isotonic calibration.

2007 Cross-Sectional Ranking: Crisis Heterogeneity

E1-EBM probability ranking by crisis type — exploratory and post-hoc



Model discriminates by mechanism

Fundamental (credit boom) crises: mean 0.695. Contagion: 0.395. No crisis: 0.325. Model trained on macro credit-cycle precursors assigns higher risk to domestically generated stress — consistent with the feature set's scope.

This analysis is explicitly exploratory and post-hoc. GBR qualitative scope failure (target_h2=0 under JST chronology). SWE true positive (Contagion ★, target_h2=1).

Hypothesis Verdicts and Four Conclusions

H1 · H2 · H3 — and what they collectively establish

H1

FinBERT improves AUPRC vs macro baseline?

M3 0.411 > M2m 0.343. CI spans zero. EBM macro-only 0.532 > M3. Algorithm dominates.

Directionally supported

H2

FinBERT feature in top-5 SHAP?

No sentiment in top-20 EBM. Macro = 73.8% SHAP. Convergence = sample correlation, not causal signal.

Not supported

H3

Quantifiable accuracy-interpretability trade-off?

Trade-off not binary. EBM leads in-sample, collapses prospectively. Flexibility = interpretability = overfitting. Same mechanism.

Supported & refined

1

More data ≠ fix

15 additional years incl. Nordic cluster does not resolve GFC generalisation. Regime discontinuity is structurally too large.

2

Interpretability ≠ robustness

EBM leads in-sample, fails most severely prospectively. Flexibility enables both interpretability and regime-specific overfitting.

3

EBM makes failure visible

Shape functions expose mechanism. Static EWS may be conceptually misaligned with structural break environments.

4

Sentiment = directional only

FinBERT gain is GFC-episode-specific. Algorithm choice explains at least as much. Multi-episode validation needed.

Implications and Close

What should change in EWS design

1

Prospective validation is mandatory

Expanding-window metrics overstate performance. Strict temporal validation must be a required evaluation step, not optional. Models failing it must not be deployed without explicit disclosure.

2

EBM for diagnostic transparency

Shape function stability analysis should be a mandatory model validation step. Regulators who can see that a model's learned mapping has shifted are better positioned to apply appropriate scepticism.

3

Ensemble for calibration

Equal-weighted averaging (Brier 0.056) is defensible for probability calibration and policy threshold-setting. It reduces discrimination (AUPRC 0.433 vs EBM 0.532) — choice depends on operational priority.

4

Sentiment needs multi-episode validation

Current evidence is hypothesis-generating. Topic-specific features (Ho et al., 2022) are more promising than polarity scores, but the topic extension found zero incremental gain.

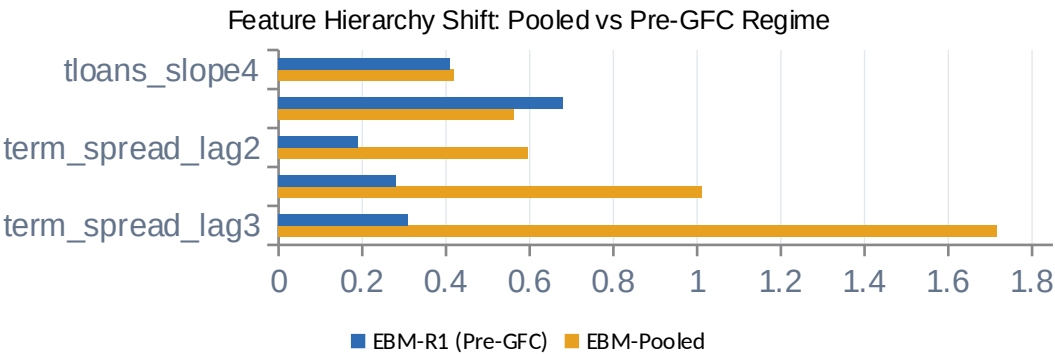
"The central limitation of EWS is the absence of a stable mapping between predictors and crisis outcomes across regimes. ML makes this visible; it cannot resolve it."

Appendix B: Exploratory Regime-Conditional Prototype

NB11 · Proof-of-concept · Confirms the binding statistical constraints

SCOPE: Exploratory extension only. Not a validated system. Demonstrates feasibility and confirms why a fully validated regime-conditional EWS cannot be implemented within the current dataset.

Regime	Period	Train positives	EBM AUROC	vs Benchmark
Pooled (baseline)	1988–2020	37	0.753 (exp-window)	Meets 0.75
EBM-Pooled (prospective)	1988–2002→2003+	6	0.410	Below 0.75 ✗
EBM-R1 (Pre-GFC)	1988–2002	6	0.410	Below 0.75 ✗
EBM-R2 (GFC, 2 positives)	2007–2010	2	0.392 (unreliable)	Insufficient data



Key finding from Appendix B

Feature hierarchy shifts between regimes: Pooled dominated by term spread at all lags; Pre-GFC dominated by CPI and debt-to-GDP. This confirms that regime-conditional modelling is conceptually motivated — but statistically constrained by crisis episode scarcity per regime.

Q&A — 6 Challenging Questions

5-minute window · Be concise · Answers in speaker notes

Q1

Split B has only 6 training positives. How can you draw conclusions from it?

Q4

Stage 2 AUROC 0.93–0.96 is inflated by GFC concentration. Are these numbers informative?

Q2

EBM 0.333 could be sample thinness, not regime shift. How do you distinguish?

Q5

Why not use a regime-switching model to directly address non-stationarity?

Q3

You cannot separate sentiment from algorithm in Stage 2. Is H1 meaningful?

Q6

SHAP already shows feature instability across time. What does EBM add?